

Labetalol: Drug information - UpToDate

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For additional information see "Labetalol: Patient drug information" and "Labetalol: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: Canada

- APO-Labetalol;
- RIVA-Labetalol;
- Tranxate

Pharmacologic Category

- Antihypertensive;
- Beta-Blocker With Alpha-Blocking Activity

Dosing: Adult

Dosage guidance:

Safety: Accumulation can occur with high-dose continuous infusions and may result in severe hypotension and bradycardia (Ref). There is limited documentation of prolonged continuous infusions (Ref).

Acute aortic syndromes/acute aortic dissection

Acute aortic syndromes/acute aortic dissection (alternative agent) (off-label use):

Note: Manage patients on an emergency basis (including operative assessment) by first controlling pain with IV opioids and heart rate (target 60 to 80 beats per minute) with a parenteral beta blocker (eg, labetalol) while targeting systolic BP of ≤ 120 mm Hg (or lowest tolerated pressure without compromising perfusion). If systolic BP remains elevated after heart rate is controlled at 60 to 80 beats per minute with beta-blockade, may consider adding an additional parenteral vasodilator (ie, nitroprusside or nicardipine). Invasive BP monitoring, preferably via arterial line, in a critical care unit is recommended for appropriate dose titration (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in select patients, while monitoring for accumulation (Ref).

Intermittent IV: Initial: 20 mg over 2 minutes followed by 20 to 80 mg every 10 minutes until target heart rate and BP are reached; may transition to continuous infusion if unable to obtain target goals (Ref).

Continuous IV infusion: Initial loading dose: 20 mg over 2 minutes (loading dose is optional if intermittent dose is used prior to continuous infusion), followed by 0.5 to 2 mg/minute; some patients may require titration up to 10 mg/minute for optimal response (Ref).

Acute ischemic stroke, BP management with reperfusion therapy

Acute ischemic stroke, BP management with reperfusion therapy (off-label use):

Note: Prior to reperfusion therapy (thrombolytic and/or mechanical thrombectomy), maintain a target BP of $\leq 185/110$ mm Hg. If BP remains $>185/110$ mm Hg, do **not** administer thrombolytic. During and 24 hours after start of thrombolytic therapy, maintain a target BP of $\leq 180/105$ mm Hg; if hypertension is refractory or diastolic BP >140 mm Hg, consider alternative therapy (Ref).

Prior to reperfusion therapy:

IV: 10 to 20 mg over 1 to 2 minutes; may repeat once (Ref).

During and after reperfusion therapy:

IV: 10 mg over 1 to 2 minutes, followed by 2 to 8 mg/minute continuous infusion (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in select patients, while monitoring for accumulation (Ref).

Hypertension, chronic/resistant

Hypertension, chronic/resistant:

Note: Not recommended for initial management of hypertension, but may be considered as additional therapy in patients who do not respond adequately to combination therapy with preferred agents (Ref).

Oral: Initial: 100 mg twice daily; may increase as needed every 2 to 3 days by 100 mg twice daily (titration increments not to exceed 200 mg twice daily) until desired response is obtained; usual dosage range: 200 to 1,200 mg/day in 2 divided doses (Ref). In patients with resistant hypertension, doses up to 1.2 to 2.4 g/day in 2 or 3 divided doses may be required; however, combination therapy is preferred over dose escalation beyond the usual dosage range (Ref).

Hypertensive emergency

Hypertensive emergency:

Note : When used for hypertensive emergency without a compelling condition, in general, reduce mean arterial blood pressure gradually by $\sim 10\%$ to 20% over the first hour. If stable, may then reduce to a goal BP of $<160/100$ mm Hg for the next 2 to 6 hours, then to 130 to 140 mm Hg over the next 24 to 48 hours to limit target organ injury (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in select patients, while monitoring for accumulation (Ref).

Intermittent IV: Initial: 10 to 20 mg over 1 to 2 minutes followed by 20 to 80 mg every 10 minutes until target blood pressure is reached; consider a continuous infusion if unable to obtain target blood pressure (Ref).

Continuous IV infusion: Initial loading dose: 10 to 20 mg over 2 minutes (loading dose is optional if intermittent dose is used prior to continuous infusion), followed by 0.5 to 2 mg/minute; some patients may require titration up to 10 mg/minute (Ref).

Hypertensive emergency in pregnancy or postpartum

Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia) (off-label use):

Note: For acute-onset, severe, persistent hypertension (eg, systolic BP ≥ 160 mm Hg or diastolic BP ≥ 110 mm Hg) (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in selected patients, while monitoring for accumulation (Ref).

Intermittent IV: Initial: 20 mg over 2 minutes; if blood pressure exceeds thresholds after 10 to 30 minutes, increase dose in increments of 20 to 40 mg every 10 to 30 minutes; maximum single dose: 80 mg. **Note:** If blood pressures remain above threshold after several intermittent doses, another agent should be used (Ref).

Continuous IV infusion: Initial loading dose: 20 mg over 2 minutes (loading dose is optional if intermittent dose is used prior to continuous infusion), followed by 0.5 to 2 mg/minute titrated to response (Ref).

Oral (alternative route): Initial: 200 mg every 60 minutes for up to 3 doses. **Note:** May use as alternative route to other ineffective oral therapy; IV therapy (intermittent or continuous infusion) may be needed for more acute treatment (Ref).

Intracerebral hemorrhage, acute, BP management

Intracerebral hemorrhage, acute, BP management (off-label use):

Note: For use in patients with mild to moderate intracerebral hemorrhage with a presenting systolic BP ≥ 150 mm Hg; acute lowering to <130 mm Hg is potentially harmful (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in select patients, while monitoring for accumulation (Ref).

Patients who present with systolic BP 150 to 220 mm Hg:

Intermittent IV: Initial: 5 to 20 mg IV push over 2 minutes. May repeat every 10 to 15 minutes; repeat doses may be increased incrementally up to 80 mg as needed based on response and tolerability until target systolic BP is reached; may consider a continuous infusion if unable to obtain target goals (Ref).

Patients who present with systolic BP >220 mm Hg (alternative agent):

Continuous IV infusion: Initial loading dose: 20 mg over 2 minutes, followed by 0.5 to 2 mg/minute; titrate to target systolic BP (Ref).

Perioperative hypertension

Perioperative hypertension:

Note: For patients with chronic hypertension prior to surgery, restart oral therapies as soon as appropriate once hemodynamically stable (Ref).

Intermittent IV: Initial: 5 to 20 mg over 2 minutes; followed by 5 to 40 mg every 5 to 10 minutes as needed until target BP is reached; may increase up to a maximum single dose of 80 mg if needed (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in selected patients, while monitoring for accumulation (Ref).

Continuous IV infusion: Initial loading dose: 5 to 20 mg IV bolus over 2 minutes (loading dose is optional if intermittent dose is used prior to continuous infusion), followed by 0.5 to 2 mg/minute continuous IV infusion titrated to target BP; some patients may require titration up to 10 mg/minute (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in selected patients, while monitoring for accumulation (Ref).

Subarachnoid hemorrhage, acute, BP management

Subarachnoid hemorrhage, acute, BP management (off-label use):

Note: Optimal therapy is not well established. Cautious use of antihypertensive therapy to decrease the risk of rebleeding, while avoiding hypotension and blood pressure variability, may be appropriate in some patients presenting with severe hypertension (Ref). Although manufacturer's labeling recommends against exceeding a cumulative IV dose of 300 mg, it may be reasonable to exceed this dose in selected patients, while monitoring for accumulation (Ref).

Intermittent IV: Initial: 10 to 20 mg over 2 minutes; followed by 20 to 80 mg every 10 to 15 minutes until systolic BP <160 mm Hg or mean arterial pressure <100 mm Hg (Ref).

Continuous IV infusion: 0.5 to 2 mg/minute titrated to response; based on very limited data (Ref).

IV to oral conversion:

Upon discontinuation of continuous IV infusion, may initiate oral dose of 200 mg followed in 6 to 12 hours with an additional dose of 200 to 400 mg; adjust dose based on response at ≥ 1 -day intervals to a range of 400 mg/day to 2.4 g/day in 2 to 3 divided doses. **Note:** For hypertension, the usual dosage range is 200 to 1,200 mg/day in 2 divided doses (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, Oral: Mild to severe impairment: No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Poorly dialyzed (manufacturer's labeling); **IV, Oral:** no supplemental dose or dosage adjustment necessary (expert opinion).

Peritoneal dialysis: Poorly dialyzed (manufacturer's labeling); **IV, Oral:** no dosage adjustment necessary (expert opinion).

CRRT: IV, Oral: No dosage adjustment necessary (expert opinion).

PIRRT (eg, sustained, low-efficiency diafiltration): IV, Oral: No dosage adjustment necessary (expert opinion).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, dosage reduction may be necessary in hepatic impairment due to decreased metabolism and increased oral bioavailability, use with caution.

Dosing: Older Adult

Refer to adult dosing.

Hypertension: Oral:

Manufacturer's labeling: Initial: 100 mg twice daily; may titrate in increments of 100 mg twice daily; usual maintenance: 100 to 200 mg twice daily

ACCF/AHA Expert Consensus recommendations: Consider lower initial doses and titrating to response (Ref)

Dosing: Pediatric

(For additional information see "Labetalol: Pediatric drug information")

Dosage guidance:

Dosing: Use care with labetalol continuous IV infusions; the rate of administration is different for pediatric patients (mg/kg/hour) versus adult patients (mg/minute).

Hypertension

Hypertension:

Acute severe hypertension with life-threatening symptoms: Limited data available:

IV (intermittent bolus): Children and Adolescents: 0.2 to 1 mg/kg/dose; maximum dose: 40 mg/dose (Ref).

Continuous IV infusion: Infants, Children, and Adolescents: 0.25 to 3 mg/kg/**hour**; initiate at lower end of range and titrate up slowly to effect (Ref). In one retrospective study in infants and children ≤ 24 months of age, reductions in blood pressure were observed at doses up to 0.59 mg/kg/hour, with little additional benefit at higher doses (Ref).

Chronic hypertension: Limited data available: Children and Adolescents: Oral: Initial: 1 to 3 mg/kg/**day** in 2 divided doses; maximum daily dose: 10 to 12 mg/kg/**day**, up to 1,200 mg/**day** (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. <1% removed by hemo- or peritoneal dialysis.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; however, dosage reduction may be necessary in hepatic impairment due to decreased metabolism and increased oral bioavailability; use with caution.

Adverse Reactions (Significant): Considerations

Bradyarrhythmias

Beta-blockers may cause **bradycardia** and **heart block**, including first-degree atrioventricular (AV) block, second-degree AV block, or complete AV block (Ref). At maintenance dosing, second- or third-degree AV block are less likely (Ref). Beta-blocking agents with intrinsic sympathomimetic activity (ISA) (eg, pindolol) may cause fewer AV conduction abnormalities than those without intrinsic sympathomimetic activity (eg, carvedilol) due to their partial agonist effects (Ref); labetalol possesses a very small degree of ISA and exhibits a slight, but irregular effect on AV conduction (Ref). In most cases (up to 72%), AV block associated with a beta-blocker will resolve upon discontinuation; however, there are reported cases of recurrent AV block and nearly 50% of patients with more severe AV block may require a permanent pacemaker (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Blockade of cardiac beta-1 adrenergic receptors results in slowed conduction and prolongation of the refractory period of the AV node. Slowing of AV conduction can lead to AV block (Ref).

Onset: Varied; one study included patients who were on a beta-blocker for more than 1 month (Ref); however, other studies noted prolongation AV nodal refractory period occurring anywhere from 1 dose to several days following treatment initiation (Ref).

Risk factors:

- Impaired AV node conduction or sinus node function (Ref)
- Concurrent use of other agents that impair AV nodal conduction (eg, nondihydropyridine calcium channel blockers, digoxin, ivabradine, select antiarrhythmic agents) (Ref)
- Older patients (Ref)

Bronchospasm

Nonselective beta-blockers (eg, labetalol) have a higher risk of **bronchospasm** compared to cardioselective beta-blockers and may lead to drug discontinuation in patients with chronic obstructive pulmonary diseases (COPD) or asthma (Ref). Use of labetalol in patients with hypertension and COPD or asthma has been associated with varying degrees of impact on FEV₁, forced expiratory flow, and/or forced vital capacity (Ref). Based on the use of other beta-blockers in this setting, concurrent use

of inhaled bronchodilators and/or corticosteroids are protective against beta-blocker-induced bronchospasm in patients with COPD or asthma (Ref). Bronchospasm is reversible upon discontinuation (Ref).

Mechanism: Dose-related; related to pharmacologic action. Beta-blocking agents can lead to airway smooth muscle constriction by antagonism of beta-2 receptors (Ref).

Onset: Varied; changes in pulmonary function tests have occurred after 1 dose (Ref).

Risk factors:

- Reactive airway disease (asthma) (Ref)

CNS effects

Beta-blockers may cause CNS effects such as **fatigue**, sleep disorder, insomnia, and vivid dreams (Ref). Lipophilic beta-blockers (including labetalol) penetrate the blood-brain barrier to a greater extent than hydrophilic beta-blockers, possibly leading to a greater incidence of CNS effects; however, other studies have refuted this theory (Ref). Sexual disorders may occur; however, patients who require beta-blocker therapy have risk factors for erectile dysfunction (eg, coronary artery disease, heart failure) (Ref). Memory impairment has been reported with the use of other lipophilic beta-blockers (eg, carvedilol); however newer data suggests carvedilol may be helpful for patients with Alzheimer disease (Ref). CNS effects generally resolve with dose reduction or discontinuation (Ref).

Mechanism: Dose-related; exact mechanism is not fully understood. Proposed mechanisms include presence of beta receptors in the brain, affinity and in some instances, inhibition of beta-blocking agents towards serotonin (5-HT) receptors in the brain (affecting mood and sleep), and beta-blocker-induced decreases in central sympathetic output (Ref). Beta-1 blockade may also impact sleep by blocking sympathetic signaling to the pineal gland, resulting in suppression of nighttime levels of melatonin (Ref). Beta-blockers may cause erectile dysfunction through decreased sympathetic nervous system output and subsequent decreases in luteinizing hormone secretion and testosterone stimulation (Ref).

Onset: Intermediate; CNS effects often occur within the first few weeks of treatment (Ref).

Risk factors:

- Higher starting doses (Ref)
- Older patients (Ref)

Potential/masking of hypoglycemia

Beta-blockers may worsen, prolong, or cause hypoglycemia (Ref). Additionally, beta-blockers may mask symptoms of hypoglycemia (tremor, irritability, palpitations), making diaphoresis the only symptom unaffected by beta-blockers (Ref). It is unclear if nonselective or selective beta-blockers are more likely to cause hypoglycemia; as data are conflicting (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Beta-blockers inhibit hepatic gluconeogenesis and glycogenolysis (Ref). Beta-blockers also reduce activation of the sympathetic nervous system, therefore masking hypoglycemic symptoms that are catecholamine-mediated (Ref).

Onset: Varied; hypoglycemia (<70 mg/dL) was reported significantly more after 24 hours of nonselective beta-blocker (ie, carvedilol) use compared to patients not on beta-blockers (Ref). In another study, episodes of severe hypoglycemia (<50 mg/dL) were reported over the course of 4 years (Ref).

Risk factors:

- Insulin-dependent diabetes (Ref)
- Type 2 diabetes mellitus (Ref)
- Patients who are hospitalized and not requiring basal insulin (Ref)

Withdrawal

Beta-blocker therapy should not be withdrawn abruptly, but gradually tapered to avoid acute tachycardia, hypertension, and/or ischemia in patients with underlying cardiovascular disease (Ref). Some studies have found an increase in propensity-adjusted mortality and cardiovascular events; however, one study did not find changes in infarct size and left ventricular function when beta-blocker was abruptly withdrawn in patients with myocardial infarction (Ref).

Mechanism: Dose-dependent; related to the pharmacologic action. Beta blockade causes upregulation of beta-receptors, enhanced receptor sensitivity, and decreased sympathetic nervous system response. Abrupt withdrawal leads to a transient sympathetic hyper-response (Ref). Another proposed mechanism involves increased platelet aggregability to epinephrine and thrombin (Ref).

Onset: Rapid/varied and transient; increases in heart rate and blood pressure appear 24 hours after abrupt withdrawal, peak after 48 hours, and subside after 7 days (Ref). In other nonselective beta-blockers (propranolol), anginal symptoms reported to begin 12 to 24 hours after discontinuation. Development of adverse reactions also reported to occur 1 to 21 days after withdrawal (Ref).

Risk factors:

- Abrupt withdrawal in chronic users (Ref)
- Past medical history of coronary artery disease (including chronic stable angina) (Ref)
- Past medical history of hypertension (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Orthostatic hypotension (IV: 58%; oral: 1%)

Gastrointestinal: Nausea ($\leq 19\%$)

Nervous system: Dizziness (1% to 16%)

1% to 10%:

Cardiovascular: Edema (oral: 1% to 2%), flushing (IV: 1%), hypotension (IV: 1%), ventricular arrhythmia (IV: 1%)

Dermatologic: Diaphoresis (IV: 4%; oral: <1%), pruritus (IV: 1%), skin rash (oral: 1%)

Gastrointestinal: Dysgeusia (1%), dyspepsia (IV: 1%; oral: 1% to 4%), vomiting (oral: $\leq 4\%$)

Genitourinary: Ejaculatory failure (oral: 1% to 5%), impotence (oral: 1% to 4%)

Hepatic: Increased serum transaminases (oral: 4%)

Nervous system: Drowsiness (IV: $\leq 3\%$), fatigue (oral: 1% to 10%) (table 1) Fatigue, headache (oral: 2%), hypoesthesia (IV: 1%), paresthesia ($\leq 7\%$), vertigo (1% to 2%), yawning (IV: $\leq 3\%$)

Table 1

Drug (Labetalol)	Comparator (Propranolol)	Comparator (Metoprolol)	Placebo	Dose	N
10%	N/A	N/A	N/A	2,400 mg/day	1
7%	N/A	N/A	N/A	1,200 mg/day	4
6%	N/A	N/A	N/A	1,600 mg/day	2
5%	12%	12%	0%	N/A	2
5%	N/A	N/A	N/A	800 mg/day	5
4%	N/A	N/A	N/A	400 mg/day	6
4%	N/A	N/A	N/A	600 mg/day	6
3%	N/A	N/A	N/A	900 mg/day	1
2%	N/A	N/A	N/A	200 mg/day	5

Drug (Labetalol)	Comparator (Propranolol)	Comparator (Metoprolol)	Placebo	Dose	N
1%	N/A	N/A	N/A	300 mg/day	1

Ophthalmic: Visual disturbance (oral: 1%)

Renal: Increased blood urea nitrogen ($\leq 8\%$), increased serum creatinine (IV: 8%)

Respiratory: Dyspnea (oral: 2%), nasal congestion (oral: 1% to 6%), wheezing (IV: 1%)

<1%: Gastrointestinal: Diarrhea

Postmarketing:

Cardiovascular: Bradycardia, heart block, heart failure, syncope

Dermatologic: Facial erythema, lichen planus (bullous), lichenoid eruption, maculopapular rash, psoriasiform eruption, transient alopecia, urticaria

Genitourinary: Acute urinary retention, difficulty in micturition, Peyronie disease

Hematologic & oncologic: Positive ANA titer

Hepatic: Cholestatic jaundice, hepatic injury, hepatic necrosis, hepatitis

Hypersensitivity: Angioedema, hypersensitivity reaction, nonimmune anaphylaxis

Immunologic: Antibody development (antimitochondrial)

Neuromuscular & skeletal: Muscle cramps, myopathy (Willis 1990), systemic lupus erythematosus

Ophthalmic: Dry eye syndrome

Respiratory: Bronchospasm

Miscellaneous: Fever

Contraindications

Hypersensitivity (eg, anaphylaxis) to labetalol or any component of the formulation; severe sinus bradycardia; heart block greater than first degree; cardiogenic shock; bronchial asthma or obstructive airway disease; decompensated heart failure (oral only); administration of non-dihydropyridine calcium channel blockers (eg, verapamil).

Canadian labeling: Additional contraindications (not in the US labeling): Sick sinus syndrome (except in patients with a functioning artificial pacemaker); state of hypoperfusion; severe peripheral arterial circulatory disorders.

Warnings/Precautions

Concerns related to adverse effects:

- Anaphylactic reactions: Use caution with history of severe anaphylaxis to allergens; patients taking beta-blockers may become more sensitive to repeated challenges. Treatment of anaphylaxis (eg, epinephrine) in patients taking beta-blockers may be ineffective or promote undesirable effects.
- Floppy iris syndrome: Intraoperative floppy iris syndrome has been observed in cataract surgery patients who were treated with α_1 -blockers. There appears to be no benefit in discontinuing α -blocker therapy prior to surgery.
- Hepatic injury: Severe hepatocellular injury has been reported (rare). The hepatic injury is usually reversible, but hepatic necrosis and death have been reported. Injury has occurred after both short- and long-term treatment and may be slowly progressive despite minimal symptomatology.

Disease-related concerns:

- Angina: Use with caution in patients suspected of having vasospastic angina.

- Heart failure (HF): Use with extreme caution in patients with compensated heart failure and monitor for a worsening of the condition.
- Hepatic impairment: Use with caution in patients with hepatic impairment; bioavailability is increased due to decreased first-pass metabolism.
- Myasthenia gravis: Use beta blockers with caution in patients with myasthenia gravis.
- Peripheral vascular disease (PVD) and Raynaud disease: Beta blockers may precipitate or aggravate symptoms of arterial insufficiency in patients with PVD and Raynaud disease; use with caution and monitor for progression of arterial obstruction.
- Pheochromocytoma: Labetalol may be effective in lowering blood pressure and relieving symptoms in patients with pheochromocytoma; however, patients may experience paradoxical hypertensive responses due to inadequate alpha-1 blockade (Manger 2002; Mazza 2014). Adequate alpha-1 blockade should be initiated prior to use of any beta-blocker in this setting; use with caution in patients with pheochromocytoma or consider alternative therapy. If possible, obtain diagnostic tests for pheochromocytoma prior to use since labetalol may spuriously cause falsely elevated levels of plasma catecholamine and urinary metanephrine (Bravo 2002; MacCarthy 1983).
- Psoriasis: Beta-blocker use has been associated with induction or exacerbation of psoriasis, but cause and effect have not been firmly established.
- Thyroid disease: Beta blockers may mask signs of hyperthyroidism (eg, tachycardia). If hyperthyroidism is suspected, carefully manage and monitor; abrupt withdrawal may exacerbate symptoms of hyperthyroidism or precipitate thyroid storm.

Special populations:

- Older adult: Bradycardia and orthostatic hypotension may be observed more frequently in older adult patients (>65 years of age); dosage reductions may be necessary.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [“Inactive” 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.
- Injection: Patients who are tilted during administration or allowed to assume the upright position within 3 hours of administration may experience symptomatic postural hypotension.

Other warnings/precautions:

- Major surgery: Chronic beta-blocker therapy should not be routinely withdrawn prior to major surgery; abrupt discontinuation may result in rebound hypertension and is potentially harmful. In beta-blocker naive patients, do not initiate therapy on the day of surgery (ACC/AHA [Jones 2025]).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as hydrochloride:

Generic: 5 mg/mL (4 mL, 20 mL, 40 mL); 100 mg/100 mL (1 mg/mL) with NaCl 0.72% (100 mL [DSC]); 200 mg/200 mL (1 mg/mL) with NaCl 0.72% (200 mL [DSC]); 200 mg/200 mL (1 mg/mL) with dextrose 5% (200 mL [DSC]); 300 mg/300 mL (1 mg/mL) in NaCl 0.72% (300 mL [DSC])

Solution, Intravenous, as hydrochloride [preservative free]:

Generic: 5 mg/mL (4 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride:

Generic: 20 mg/4 mL (4 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride [preservative free]:

Generic: 10 mg/2 mL (2 mL [DSC])

Tablet, Oral, as hydrochloride:

Generic: 100 mg, 200 mg, 300 mg, 400 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Labetalol HCl Intravenous)

5 mg/mL (per mL): \$0.13 - \$0.48

Solution Prefilled Syringe (Labetalol HCl Intravenous)

20 mg/4 mL (per mL): \$2.72

Tablets (Labetalol HCl Oral)

100 mg (per each): \$0.17 - \$0.67

200 mg (per each): \$0.26 - \$1.14

300 mg (per each): \$0.35 - \$1.15

400 mg (per each): \$2.27

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as hydrochloride:

Generic: 5 mg/mL (20 mL, 40 mL)

Tablet, Oral, as hydrochloride:

Trandate: 100 mg [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Trandate: 200 mg

Generic: 100 mg, 200 mg

Extemporaneous Preparations

40 mg/mL Oral Suspension (ASHP Oral Suspension) (Ref)

A 40 mg/mL labetalol hydrochloride oral suspension may be made with tablets and one of three different vehicles (cherry syrup, a 1:1 mixture of Ora-Sweet and Ora-Plus, or a 1:1 mixture of Ora-Sweet SF and Ora-Plus). Crush sixteen 300 mg tablets in a mortar and reduce to a fine powder. Add 20 mL of the chosen vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 120 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of vehicle sufficient to make 120 mL. Label "shake well" and "protect from light." Stable for 60 days when stored in amber plastic prescription bottles in the dark at room temperature or refrigerated (Ref).

10 mg/mL Oral Suspension

A 10 mg/mL labetalol hydrochloride oral suspension may be made with tablets and simple syrup. Crush twelve 100 mg tablets in a mortar and reduce to a fine powder. Add a small quantity of simple syrup and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 120 mL; transfer to a calibrated bottle, rinse mortar with simple syrup, and add sufficient quantity to make 120 mL. Label "shake well" and "refrigerate." Stable for 4 weeks when stored under refrigeration (preferred) or at room temperature (Ref).

Administration: Adult

Oral: Administer without regard to food; however, the absolute bioavailability of labetalol is increased when administered with food. Administer in a consistent manner with regards to meals.

Parenteral: Bolus dose of 10 mg may be administered IV push at a rate of 10 mg/minute; bolus doses of ≥ 20 mg may be administered IV push over 2 minutes; may follow with continuous IV infusion if clinically indicated (Ref). Orthostatic hypotension may occur with IV administration; patient should remain supine during and for up to 3 hours after IV administration; patient's ability to tolerate an upright position should be established before permitting ambulation or assuming an upright position.

Preparation for Administration: Adult

Note: Premixed solutions are available.

Parenteral: Continuous IV infusion: Further dilute in a compatible solution (eg, D5W, NS, D5NS, LR) to a concentration of 1 mg/mL; concentrations up to 3.75 mg/mL have been shown to be stable (Ref). For patients with fluid overload, may also prepare a 5 mg/mL solution for continuous infusion by adding contents from 5 mg/mL vial(s) to a compatible container with no further dilution required (Ref).

Usual Infusion Concentrations: Adult

Note: Premixed solutions are available.

IV infusion: 200 mg in 200 mL (concentration: 1 mg/mL) **or** 400 mg in 200 mL (concentration: 2 mg/mL) **or** 500 mg in 250 mL (concentration: 2 mg/mL) of D5W.

Administration: Pediatric

Oral: Administer without regard to food; however, the absolute bioavailability of labetalol is increased when administered with food. Administer in a consistent manner with regards to meals.

Parenteral:

IV bolus: Based on adult information, may administer undiluted over 2 minutes; maximum: 10 mg/minute.

Continuous IV infusion: Administer as a continuous IV infusion with the use of an infusion pump.

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Vials: Further dilute in a compatible solution (eg, D5W, D5NS, NS, LR) to a concentration of 0.67 to 1 mg/mL; premixed IV solutions are available (1 mg/mL). ASHP recommends standard concentrations of 1 mg/mL and 5 mg/mL (undiluted) in pediatric patients (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 1 mg/mL, 5 mg/mL.

Storage/Stability

Tablets: Store at 2°C to 30°C (36°F to 86°F).

Injectable (prefilled syringes, premixed bags, vials): Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C to 30°C (59°F to 86°F). Do not freeze. Protect from light. For premixed bags, do not remove overwrap until ready to use and discard any remaining solution 24 hours after infusion started.

Parenteral admixture: Stability of parenteral admixture at room temperature (25°C [77°F]) and refrigeration temperature (4°C [39°F]): 3 days (Yuen 1983).

Use: Labeled Indications

Hypertension: Management of hypertension (IV indicated for severe hypertension only [eg, hypertensive emergencies]).

Use: Off-Label: Adult

Acute aortic syndromes/Acute aortic dissection; Acute ischemic stroke, BP management with reperfusion therapy; Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia); Intracerebral hemorrhage, acute, blood pressure management; Subarachnoid hemorrhage, acute, blood pressure management

Medication Safety Issues

Sound-alike/look-alike issues:

Labetalol may be confused with betaxolol, LaMictal, lamoTRigine, Lipitor

Normodyne may be confused with Norpramin

Tranxate may be confused with traMADol, Trental

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (adrenergic antagonist, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Older Adult: High-Risk Medication:

Labetalol is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age). Some disease states of concern include bradycardia, heart block, and severe symptomatic aortic stenosis. Use is not recommended as monotherapy for the treatment of uncomplicated hypertension (O'Mahony 2023).

Administration issues:

Significant differences exist between oral and IV dosing. Use caution when converting from one route of administration to another.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acetylcholinesterase Inhibitors: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alpha2-Agonists: Beta-Blockers may increase rebound hypertensive effects of Alpha2-Agonists. This effect can occur when the Alpha2-Agonist is abruptly withdrawn. Alpha2-Agonists may increase AV-blocking effects of Beta-Blockers. Sinus node dysfunction may also be enhanced. Management: Closely monitor heart rate during treatment with a beta blocker and clonidine. Withdraw beta blockers several days before clonidine withdrawal when possible, and monitor blood pressure closely. Recommendations for other alpha2-agonists are unavailable. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amiodarone: May increase bradycardic effects of Beta-Blockers. Possibly to the point of cardiac arrest. Amiodarone may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Antidiabetic Agents: Beta-Blockers (Nonselective) may increase hypoglycemic effects of Antidiabetic Agents. Beta-Blockers (Nonselective) may increase adverse/toxic effects of Antidiabetic Agents. Specifically, beta-blockers may mask the hypoglycemic symptoms of antidiabetic agents. *Risk C: Monitor*

Antipsychotic Agents (Phenothiazines): May increase hypotensive effects of Beta-Blockers. Beta-Blockers may decrease metabolism of Antipsychotic Agents (Phenothiazines). Antipsychotic Agents (Phenothiazines) may decrease metabolism of Beta-Blockers. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta2-Agonists: Beta-Blockers (Nonselective) may decrease bronchodilatory effects of Beta2-Agonists. Management: Patients with asthma or COPD receiving beta-2 agonists should not normally be treated with non-selective beta-blockers. If used concomitantly, monitor closely for diminished bronchodilatory effects of the beta2-agonist. *Risk D: Consider Therapy Modification*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Cafedrine: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Cafedrine. *Risk C: Monitor*

Cannabis: Beta-Blockers may increase adverse/toxic effects of Cannabis. Specifically, the risk of hypoglycemia may be increased. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Cholinergic Agonists: Beta-Blockers may increase adverse/toxic effects of Cholinergic Agonists. Of particular concern are the potential for cardiac conduction abnormalities and bronchoconstriction. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Dipyridamole: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Disopyramide: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may increase negative inotropic effects of Disopyramide. *Risk C: Monitor*

DOBUtamine: Beta-Blockers may decrease therapeutic effects of DOBUtamine. *Risk C: Monitor*

Dronedarone: May increase bradycardic effects of Beta-Blockers. Dronedarone may increase serum concentrations of Beta-Blockers. This likely applies only to those agents that are metabolized by CYP2D6. Management: Use lower initial beta-blocker doses; adequate tolerance of the combination, based on ECG findings, should be confirmed prior to any increase in beta-blocker dose. Increase monitoring for clinical response and adverse effects. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

EPHEDrine (Systemic): Beta-Blockers may decrease therapeutic effects of EPHEDrine (Systemic). *Risk C: Monitor*

EPINEPHrine (Systemic): Beta-Blockers (with Alpha-Blocking Properties) may decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): Beta-Blockers may increase vasoconstricting effects of Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates). *Risk C: Monitor*

Etilefrine: Beta-Blockers may decrease therapeutic effects of Etilefrine. Etilefrine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Etofylline: Beta-Blockers may decrease therapeutic effects of Etofylline. *Risk X: Avoid*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Grass Pollen Allergen Extract (5 Grass Extract): Beta-Blockers may increase adverse/toxic effects of Grass Pollen Allergen Extract (5 Grass Extract). More specifically, Beta-Blockers may inhibit the ability to effectively treat severe allergic reactions to Grass Pollen Allergen Extract (5 Grass Extract) with epinephrine. Some other effects of epinephrine may be unaffected or even enhanced (e.g., vasoconstriction) during treatment with Beta-Blockers. Management: Consider alternatives to either grass pollen allergen extract (5 grass extract) or beta-blockers in patients with indications for both agents. Canadian product labeling specifically lists this combination as contraindicated. *Risk D: Consider Therapy Modification*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Insulin: Beta-Blockers (Nonselective) may increase hypoglycemic effects of Insulin. Beta-Blockers (Nonselective) may decrease therapeutic effects of Insulin. *Risk C: Monitor*

Iobenguane Radiopharmaceutical Products: Labetalol may decrease therapeutic effects of Iobenguane Radiopharmaceutical Products. Management: Discontinue all drugs that may inhibit or interfere with catecholamine transport or uptake for at least 5 biological half-lives before Iobenguane administration. Do not administer labetalol until at least 7 days after each Iobenguane dose. *Risk X: Avoid*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Isoproterenol: Beta-Blockers may decrease therapeutic effects of Isoproterenol. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mavacamten: Beta-Blockers may increase adverse/toxic effects of Mavacamten. Specifically, negative inotropic effects may be increased. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methacholine: Beta-Blockers may increase adverse/toxic effects of Methacholine. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mivacurium: Beta-Blockers may increase therapeutic effects of Mivacurium. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

NIFEdipine (Systemic): May increase hypotensive effects of Beta-Blockers. NIFEdipine (Systemic) may increase negative inotropic effects of Beta-Blockers. *Risk C: Monitor*

NIFEdipine (Topical): May increase adverse/toxic effects of Beta-Blockers. *Risk C: Monitor*

Nitrendipine: May increase therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May decrease therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease antihypertensive effects of Beta-Blockers. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Opipramol: Beta-Blockers may increase serum concentrations of Opipramol. Opipramol may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reserpine: May increase hypotensive effects of Beta-Blockers. Reserpine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Rivastigmine: May increase bradycardic effects of Beta-Blockers. Management: Consider alternatives to the use of rivastigmine and beta-blockers when possible, as product labeling states that the concomitant use of these agents is not recommended. If combined, monitor for bradycardia and syncope. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Succinylcholine: Beta-Blockers may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Sulfonylureas: Beta-Blockers (Nonselective) may increase hypoglycemic effects of Sulfonylureas. Beta-Blockers (Nonselective) may decrease therapeutic effects of Sulfonylureas. *Risk C: Monitor*

Tasimelteon: Beta-Blockers may decrease therapeutic effects of Tasimelteon. Management: Consider avoiding nighttime administration of beta-blockers during tasimelteon therapy due to the potential for reduced tasimelteon efficacy. *Risk D: Consider Therapy Modification*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Theodrenaline: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Theodrenaline. *Risk C: Monitor*

Theophylline Derivatives: Beta-Blockers (Nonselective) may decrease bronchodilatory effects of Theophylline Derivatives. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

White Birch Allergen Extract: Beta-Blockers may increase adverse/toxic effects of White Birch Allergen Extract. Specifically, beta-blockers may reduce the effectiveness of beta-agonists that may be required to treat systemic reactions to white birch allergen extract. *Risk X: Avoid*

Zuclopenthixol: May increase or decrease antihypertensive effects of Beta-Blockers. *Risk X: Avoid*

Food Interactions

Labetalol serum concentrations may be increased if taken with food. Management: Administer with food.

Reproductive Considerations

Medications considered acceptable for the treatment of chronic hypertension during pregnancy may generally be continued in patients trying to conceive. Labetalol is one of the medications patients can transition to if currently taking an agent that is ineffective or should be avoided during pregnancy (ACC/AHA [Whelton 2018]; ACOG 2019; NICE 2019).

Erectile dysfunction and inhibition of sperm motility are noted in product labeling following use of beta-blockers, including labetalol. As a class, outcomes from available studies evaluating beta-blockers and sexual dysfunction are inconsistent, and the negative effects on erectile function are considered controversial. A clear relationship between use of beta-blockers and erectile dysfunction has not been established. Hypertension itself is associated with erectile dysfunction. Patients on a beta-blocker presenting with sexual dysfunction should be evaluated for underlying disease (Farmakis 2021; Levine 2012; Semet 2017; Terentes-Printzios 2022; Viigimaa 2020).

Pregnancy Considerations

Labetalol crosses the placenta.

Exposure to beta-blockers during pregnancy may increase the risk for adverse events in the neonate. If maternal use of a beta-blocker is needed, monitor fetal growth during pregnancy; monitor the newborn for 48 hours after delivery for bradycardia, hypoglycemia, and respiratory depression (ESC [Regitz-Zagrosek 2018]).

Chronic maternal hypertension is also associated with adverse events in the fetus/infant. Chronic maternal hypertension may increase the risk of birth defects, low birth weight, premature delivery, stillbirth, and neonatal death. Actual fetal/neonatal risks may be related to the duration and severity of maternal hypertension. Untreated chronic hypertension may also increase the risks of adverse maternal outcomes, including gestational diabetes, preeclampsia, delivery complications, stroke, and myocardial infarction (ACOG 2019).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of labetalol may be altered (Mulrenin 2021; van de Vusse 2022). Dose adjustments may be required; close monitoring is recommended (Mulrenin 2021; SOGC [Magee 2022]).

Patients with preexisting hypertension may continue their medication during pregnancy unless contraindications exist (ESC [Regitz-Zagrosek 2018]). When treatment for hypertension is initiated dur-

ing pregnancy, oral labetalol is one of the preferred agents (ACOG 2019; ESC [Cífková 2020]; ESC [Regitz-Zagrosek 2018]; SOGC [Magee 2022]). IV labetalol is recommended for use in the management of hypertensive emergencies in pregnant or postpartum patients (including acute-onset severe hypertension in preeclampsia/eclampsia) (ACOG 2020a; SOGC [Magee 2022]).

Breastfeeding Considerations

Labetalol is present in breast milk.

Data related to the presence of labetalol in breast milk are available following maternal administration of oral labetalol 300 mg twice daily to a woman for preeclampsia. Delivery was at 26 weeks' gestation; breast milk concentrations ~8 days postpartum were 0.71 mcg/mL (Mirpuri 2008). Using a milk concentration of 0.71 mcg/mL, the estimated daily infant dose via breast milk is 0.1065 mg/kg/day providing a relative infant dose (RID) of 3.6% compared to an infant therapeutic dose of 3 mg/kg/day. In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

Breast milk concentrations have not been found to correspond to maternal serum concentrations (Lunell 1985; Michael 1979). Peak breast milk concentrations are noted to occur 1 to 3 hours after the maternal dose (Lunell 1985).

Asymptomatic bradycardia was observed in a preterm infant exclusively breastfed following maternal use of labetalol (Mirpuri 2008). Nipple pain and Raynaud phenomenon of the nipple has been associated with labetalol in case reports (Rolfes 2014).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Labetalol is considered compatible for use in patients who are breastfeeding (ACOG 2019; ESC [Cífková 2020]; SOGC [Magee 2022]).

Monitoring Parameters

BP (especially with initial dosing and dosing increases), heart rate, continuous cardiac monitoring recommended for IV administration (consult individual institutional policies and procedures); periodic LFTs if clinically indicated; mental alertness; signs and symptoms of bronchospasm in patients with existing bronchospastic disease; serum glucose (in patients with diabetes).

Hypertension, acute in pregnancy (hypertensive emergency/urgency): Once target BP is achieved, monitor every 10 minutes for the first hour, then every 15 minutes for 1 hour, then every 30 minutes for 1 hour, then every hour for 4 hours (ACOG 2020b).

Reference Range

BP goal: May vary depending on clinical conditions, different clinical practice guidelines, and expert opinion. Refer to "Clinical Practice Guidelines" for specific treatment goals.

Mechanism of Action

Blocks α_1 -, β_1 -, and β_2 -adrenergic receptor sites; elevated renins are reduced. The ratios of alpha- to beta-blockade differ depending on the route of administration estimated to be 1:3 (oral) and 1:7 (IV) (Goa 1989).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: 20 minutes to 2 hours (McNeil 1984); IV: Within 5 minutes (Goa 1989)

Peak effect: Oral: 2 to 4 hours; IV: 5 to 15 minutes (Goa 1989)

Duration: Blood pressure response:

Oral: 8 to 12 hours (dose dependent)

IV: Average: 16 to 18 hours (dose dependent)

Absorption: Complete

Distribution: Apparent V_d : Adults: 2.5 to 15.7 L/kg (Goa 1989)

Protein binding: ~50%

Metabolism: Hepatic, primarily via glucuronide conjugation; extensive first-pass effect

Bioavailability: Oral: 25%; increased with liver disease, elderly, and concurrent cimetidine

Half-life elimination: Oral: 6 to 8 hours; IV: ~5.5 hours

Time to peak, plasma: Oral: 1 to 2 hours

Excretion: Urine (55% to 60% as glucuronide conjugates, <5% as unchanged drug [Goa 1989]); feces (12% to 27% as metabolites) (Goa 1989)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Elimination is reduced in elderly patients.

Patient Counseling Points

(For additional information see "Labetalol: Patient drug information")

What is this drug used for?

- It is used to treat high blood pressure.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Feeling dizzy, tired, or weak
- Upset stomach
- Tingling of the scalp
- Stuffy nose

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Liver problems like dark urine, feeling tired, not hungry, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Very bad dizziness or passing out
- Slow heartbeat
- A burning, numbness, or tingling feeling that is not normal
- Not able to get or keep an erection
- Heart failure like shortness of breath, a big weight gain, a heartbeat that is not normal, or swelling in the arms or legs that is new or worse
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT

include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Biascor | Blocamine | Gravidac | Labetalol | Labetalol richet;
- (AT) Austria: Trandate;
- (AU) Australia: Labetalol sxp | Presolol | Trandate;
- (BD) Bangladesh: Labecard | Labegest | Labeta;
- (BE) Belgium: Trandate;
- (CH) Switzerland: Trandate;
- (CL) Chile: Labetalol | Trandate | Zeverotens;
- (CN) China: Labetalol;
- (CO) Colombia: Diblorece | Labetalol | Labetalol clorhidrato | Labetalol HCL | Presequil | Telabelol | Trandate;
- (CR) Costa Rica: Biascor;
- (CZ) Czech Republic: Labetalol | Trandate;
- (EC) Ecuador: Biascor | Diblorece | Labetalol;
- (EE) Estonia: Labetalol | Labetalolo s.a.l.f. | Trandate;
- (EG) Egypt: Labipress;
- (ES) Spain: Trandate;
- (FI) Finland: Albetol | Labetalol s.a.l.f. | Trandate;
- (FR) France: Labetalol Dci | Labetalol stragen | Trandate;
- (GB) United Kingdom: Labetalol | Labetalol astec | Labetalol cox | Labetalol dc | Labetalol kent | Labetalol sandoz | Labrocol | Trandate;
- (GR) Greece: Trandate;
- (HK) Hong Kong: Labetalol HCL | Trandate;
- (HU) Hungary: Trandate;
- (ID) Indonesia: Trandate;
- (IE) Ireland: Trandate;
- (IL) Israel: Trandate;
- (IN) India: Ab lol | Alphadopa I | Culol | Eubet | Evabet | Gravidol | Gynapres L | Labebet | Labepure | Labetamac | Labetol | Labil | Lablol | Lebatens | Lexol | Lobet | Lopih | Pregnasafe | Tiplab;
- (IT) Italy: Amipress | Ipolab | Labetalolo s.a.l.f. | Trandate;
- (JP) Japan: Amitalol | Ascool | Labestonin | Labetalol hydrochloride towa | Resporito | Trandate | Trandate glaxosmithkline | Trandate takeda;
- (KR) Korea, Republic of: Betasin | Labesin | Trandate;
- (KW) Kuwait: Labetalol | Labetalol HCL | Trandate;
- (LT) Lithuania: Coreton | Labetalol | Libetor | Norbetol | Trandate | Trandate aspen;
- (LU) Luxembourg: Trandate;
- (LV) Latvia: Coreton | Trandate;
- (MY) Malaysia: Labetalol | Labetalol pharmaniaga | Pharmaniaga labetalol hydrochloride | Tolbetol | Trandate | Trantalol;
- (NG) Nigeria: Labetalol HCL;
- (NL) Netherlands: Labetalol | Labetalol B-Medical | Labetalol HCL | Labetalol hcl Alpharma | Labetalol Hcl CF | Labetalol Hcl Katwijk | Labetalol Hcl Merck | Labetalol hcl PCH | Labetalol Hcl Sandoz | Trandate;
- (NO) Norway: Labetalol bedford | Labetalol s.a.l.f. | Trandate;

- (NZ) New Zealand: Hybloc | Labetalol | Presolol | Trandate;
- (PA) Panama: Biascor | Clorhidrato de labetalol;
- (PE) Peru: Betalopharm | Labetalol;
- (PH) Philippines: Abetol;
- (PK) Pakistan: Labetalol | Labetol | Letol;
- (PL) Poland: Coreton | Labetalol HCL | Labil | Lablol | Pressocard | Trandate;
- (PR) Puerto Rico: Labetalol | Labetalol HCL | Normodyne | Trandate;
- (PT) Portugal: Labetalol pharmakern | Trandate;
- (PY) Paraguay: Biascor;
- (QA) Qatar: Labever | Trandate;
- (RO) Romania: Trandate;
- (SA) Saudi Arabia: Trandate;
- (SE) Sweden: Labetalol s.a.l.f | Trandate;
- (SG) Singapore: Trandate | Trantalol;
- (SI) Slovenia: Trandate;
- (SK) Slovakia: Trandate;
- (SR) Suriname: Labetalol;
- (TH) Thailand: Avexa | Labesin;
- (TN) Tunisia: Trandate;
- (TR) Turkey: Trandate;
- (TW) Taiwan: Abetol | Chenday | Labedin | Labenon | Labtal | Latol | Lavocohere | Presolol | Trandate | Tranmin;
- (UY) Uruguay: Biascor | Blocamine | Labetalol;
- (VE) Venezuela, Bolivarian Republic of: Trandate;
- (ZA) South Africa: Adco labetalol | Trandate

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